

Model	#Param	PPL
Inan et al. (2016) - Tied Variational LSTM	24M	73.2
Zilly et al. (2016) - Variational RHN	23M	65.4
Zoph and Le (2016) - NAS Cell	25M	64.0
Merity et al. (2017) - AWD-LSTM	24M	58.8
Pham et al. (2018) - Efficient NAS	24M	58.6
Liu et al. (2018) - Differentiable NAS	23M	56.1
Yang et al. (2017) - AWD-LSTM-MoS	22M	55.97
Melis et al. (2018) - Dropout tuning	24M	55.3
Ours - Transformer-XL	24M	54.52
Merity et al. (2017) - AWD-LSTM+Finetune [†]	24M	57.3
Yang et al. (2017) - MoS+Finetune [†]	22M	54.44

Table 5: Comparison with state-of-the-art results on Penn Treebank. [†] indicates using two-step finetuning.

One Billion Word does not preserve any long-term dependency because sentences have been shuffled. Consequently, this dataset mainly tests the ability of modeling only short-term dependency. The comparison between Transformer-XL and the other methods is shown in Table 4. Although Transformer-XL is mainly designed to better capture longer-term dependency, it dramatically improves the single-model SoTA from 23.7 to 21.8. Specifically, Transformer-XL significantly outperforms a contemporary method using vanilla Transformers (Baevski and Auli, 2018), suggesting the advantage of Transformer-XL is generalizable to modeling short sequences.

We also report the results on word-level Penn Treebank in Table 5. Similar to AWD-LSTM (Merity et al., 2017), we apply variational dropout and weight average to Transformer-XL. With proper regularization, Transformer-XL achieves a new SoTA result among models without two-step finetuning. Penn Treebank has only 1M training tokens, which implies that Transformer-XL also generalizes well even on small datasets.

4.2 Ablation Study

We conduct two sets of ablation studies to examine the effects of two proposed techniques used in Transformer-XL: the recurrence mechanism and the new positional encoding scheme.

The first study is performed on WikiText-103, which requires modeling long-term dependency. The results are reported in Table 6. Among the compared encoding schemes, Shaw et al. (2018) is relative, while Vaswani et al. (2017) and Al-Rfou et al. (2018) are absolute. “Full” and “half” losses refer to applying a cross entropy loss to all or the recent half positions in the segment. We found

that absolute encodings only work well with half losses because half losses exclude positions with very short attention lengths during training for better generalization. Table 6 shows that both the recurrence mechanism and our encoding scheme are necessary to achieve the best performance, as well as generalizing to longer attention sequences during evaluation time. Although the backpropagation length during training is only 128, with the two techniques the attention length can be increased to 640 at test time. In the standard setting with 151M parameters, the perplexity decreases as the attention length increases.

Since the recurrence mechanism costs additional memory, we also compare Transformer-XL with baselines under the same GPU memory constraints. As shown in Table 10 in Appendix A, despite using a shorter backpropagation length, Transformer-XL remains superior to the baselines.

The second study targets at isolating the effects of resolving the context fragmentation problem from the benefit of capturing longer context length. In order to achieve this goal, we deliberately choose a dataset that does not require long-term dependency, so that any improvement from establishing the recurrence can be attributed to solving the context fragmentation. Specifically, we perform this controlled experiment on the One Billion Word dataset, which can only benefit from removing the context fragmentation. We train a 20-layer Transformer-XL with ~ 0.3 B parameters for 400K steps. As shown in Table 7, using segment-level recurrence substantially improves performance even when long-term dependency is not needed, which is consistent with our previous discussion that the recurrence mechanism resolves the context fragmentation problem. Moreover, our relative positional encodings is also superior to Shaw et al. (2018) on short sequences.

4.3 Relative Effective Context Length

Khandelwal et al. (2018) proposed a method to evaluate the *Effective Context Length* (ECL) of a sequence model. ECL is the longest length to which increasing the context span would lead to a gain more than a threshold. However, ECL ignores the fact that it is harder to get improvement when a model already achieves a lower perplexity using only a shorter context, and thus it is not suitable for fair comparison among multiple models. We instead propose a new metric